

I. Fourier Transform of a Distribution.

Fact: $x(t)$ is Schwartz if and only if its Fourier transf.
is Schwartz.

Defn: Suppose $T(t)$ is a distribution associated with a sequence of Schwartz signals $f_1, f_2, \dots$.

$$\begin{aligned}\langle T, x \rangle &= \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} x(t) f_n(t) dt \\ &= \int_{-\infty}^{\infty} T(t) \cdot x(t) dt.\end{aligned}$$

The Fourier transf. of $T(t)$ is a distribution $\hat{T}(\omega)$ associated with the sequence of Schwartz signals $F_1(\omega), F_2(\omega), \dots$.

Notation: $\langle \hat{T}(\omega), X(\omega) \rangle$ or $\int_{-\infty}^{\infty} \hat{T}(\omega) X(\omega) d\omega$

$$\triangleq \lim_{n \rightarrow \infty} \left[\int_{-\infty}^{\infty} F_n(\omega) X(\omega) d\omega \right].$$

Remarks:

1. Every distribution T has a Fourier transform and this Fourier transform is invertible.
2. We denote this as:

$$T(t) \xleftrightarrow{F} \hat{T}(\omega).$$

3. If $f(t)$ is a function that is absolutely integrable :

$$(a) \langle f, \chi \rangle = \int_t f(t) \chi(t) dt \rightarrow f \text{ viewed as a distribution}$$

$$(b) F(\omega) = \int_t f(t) \cdot e^{j\omega t} dt \quad \& \quad \omega \text{ is well defined.}$$

If f is viewed as a distribution its Fourier transform is $F(\omega)$

↑
(Fourier transf. when f is viewed as a signal).

II. Properties of Fourier Transform of Distribution.

$$0. \quad T_1 \xleftrightarrow{F} \hat{T}_1, \quad T_2 \xleftrightarrow{F} \hat{T}_2, \quad \alpha_1, \alpha_2 \in \mathbb{C}$$

$$\alpha_1 T_1 + \alpha_2 T_2 \xleftrightarrow{F} \alpha_1 \hat{T}_1 + \alpha_2 \hat{T}_2.$$

1. Relation between time and frequency domain distributions:

$$\text{Suppose } T \xleftrightarrow{F} \hat{T} \quad (\text{as distribution})$$

$$\text{and } \chi \xleftrightarrow{F} X \quad (\text{as signals})$$

and if χ is Schwartz or any other well behaved signal:

$$\langle \hat{T}, X \rangle = 2\pi \cdot \langle T(t), \chi(-t) \rangle$$

$$\int_{-\infty}^{+\infty} \hat{T}(\omega) X(\omega) d\omega = 2\pi \int_{-\infty}^{+\infty} T(t) x(-t) dt.$$

Example:

① Suppose $T(t) = \delta(t)$, what is $\hat{T}(\omega)$?

$$\int_{-\infty}^{+\infty} \hat{T}(\omega) X(\omega) d\omega = 2\pi \int_{-\infty}^{+\infty} T(t) \cdot x(-t) dt$$

$$= 2\pi \int_{-\infty}^{+\infty} \delta(t) x(-t) dt$$

$$= 2\pi x(x(-t)) \Big|_{t=0}$$

$$= 2\pi x(0).$$

Since x, X are Schwartz, Fourier inversion theorem works:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) \cdot e^{j\omega t} d\omega.$$

$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) d\omega.$$

$$\therefore \int_{-\infty}^{+\infty} \hat{T}(\omega) X(\omega) d\omega = 2\pi \cdot x \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\omega) \cdot 1 \cdot d\omega$$

$$= \int_{-\infty}^{+\infty} X(\omega) \cdot 1 d\omega.$$

$$\Rightarrow \langle \hat{T}(\omega), X \rangle = \langle 1, X \rangle \text{ for any } X.$$

$$\therefore \hat{T}(\omega) = 1.$$

$$\therefore \delta(t) \leftrightarrow 1.$$

$$2. \quad T(t) = 1, \quad \hat{T}(\omega) = ??$$

Use the formula:

$$\langle \hat{T}, X \rangle = 2\pi \langle T, x(-t) \rangle$$

$$= 2\pi \int_{t=-\infty}^{+\infty} 1 \cdot x(-t) dt$$

$$= 2\pi \int_{t=-\infty}^{+\infty} x(t) \cdot dt$$

$$= 2\pi \int_{-\infty}^{+\infty} x(t) \cdot e^{j\omega t} dt \Big|_{\omega=0}$$

$$= 2\pi X(\omega) \Big|_{\omega=0}$$

$$= 2\pi X(0).$$

$$\langle \hat{T}, X \rangle = 2\pi X(0) = 2\pi \langle \delta(\omega), X \rangle.$$

$$\Rightarrow \hat{T}(\omega) = 2\pi \delta(\omega).$$

$$\Rightarrow \quad \underline{1} \xleftrightarrow{F} 2\pi \delta(\omega) \quad \Bigg\} \quad \delta(t) \xleftrightarrow{F} 1.$$

$$2. \quad \underline{\text{Duality}}: \quad \text{If } T(t) \xleftrightarrow{F} \hat{T}(\omega)$$

then: $\hat{T}(t) \xleftrightarrow{F} 2\pi \cdot T(-\omega).$

Example: As signal: $\pi(t) \xleftrightarrow{F} \text{sinc}\left(\frac{\omega}{2\pi}\right)$

$\Rightarrow$, As distribution: $\pi(t) \xleftrightarrow{F} \text{sinc}\left(\frac{\omega}{2\pi}\right)$

Apply duality: $\text{sinc}\left(t/2\pi\right) \xleftrightarrow{F} 2\pi \cdot \pi(-\omega)$
 $= 2\pi \cdot \pi(\omega).$

3. Time / Frequency Scaling:

If $T(t) \xleftrightarrow{F} \hat{T}(\omega)$ then if $a \in \mathbb{R}$ and $a \neq 0$:

$$T(at) \xleftrightarrow{F} \frac{1}{|a|} \hat{T}\left(\frac{\omega}{a}\right).$$

See: $T(-t) \xleftrightarrow{F} \hat{T}(-\omega).$

4. Time and Frequency Shifting:

If $T(t) \xleftrightarrow{F} \hat{T}(\omega)$ then:

(i) $T(t-t_0) \xleftrightarrow{F} \hat{T}(\omega) \cdot e^{-j\omega t_0}$

(ii) $T(t) \cdot e^{j\omega_0 t} \xleftrightarrow{F} \hat{T}(\omega - \omega_0).$

5. Derivative Theorems:

(i) $T'(t) \xleftrightarrow{F} j\omega \cdot \hat{T}(\omega)$

(ii) $-jt \cdot T(t) \xleftrightarrow{F} \hat{T}'(\omega).$

6. Convolution and multiplication Properties:

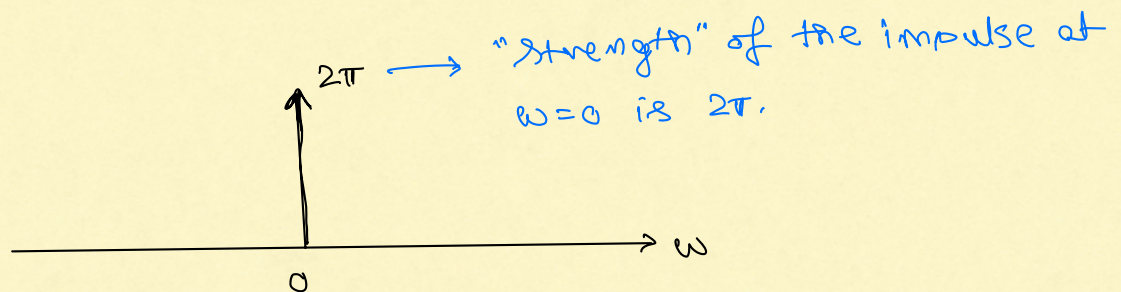
If x is Schwartz (or any well behaved signal)

$$(i) \quad x(t) * T(t) \xleftrightarrow{F} X(\omega) \cdot \hat{T}(\omega)$$

$$(ii) \quad x(t) \cdot T(t) \xleftrightarrow{F} \frac{1}{2\pi} [X(\omega) * \hat{T}(\omega)].$$

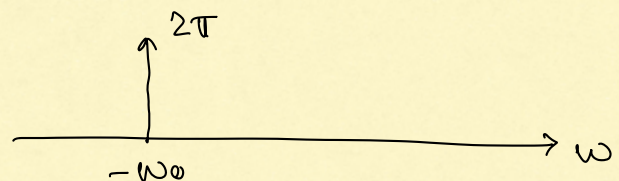
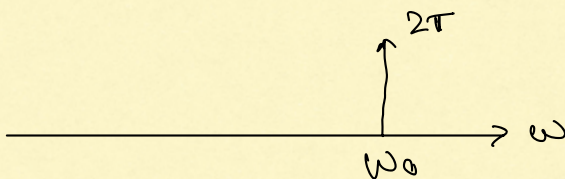
Examples:

1. We know that $1 \xleftrightarrow{F} 2\pi\delta(\omega)$



Then, $e^{j\omega_0 t} \xleftrightarrow{F} 2\pi\delta(\omega - \omega_0)$ ($\because$ frequency shifting)

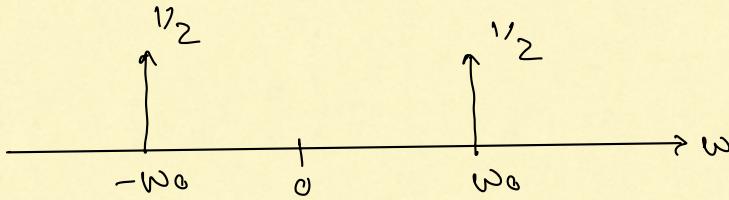
$$e^{-j\omega_0 t} \xleftrightarrow{F} 2\pi\delta(\omega + \omega_0).$$



$$\cos \omega_0 t = (e^{j\omega_0 t} + e^{-j\omega_0 t}) / 2$$

$$\xleftrightarrow{F} \frac{1}{2} [2\pi\delta(\omega - \omega_0) + 2\pi\delta(\omega + \omega_0)] \quad (\text{linearity property})$$

$$= \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$



similarly, $\sin \omega_0 t \xleftrightarrow{F} \frac{\pi}{j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$

2. We know that: $\delta(t) \xleftrightarrow{F} 1$

$\Rightarrow \delta(t - t_0) \xleftrightarrow{F} e^{-j\omega t_0}$ (time shifting property).

Multipath channel:

$\sum_{k=1}^K a_k \delta(t - t_k) \xleftrightarrow{F} \sum_{k=1}^K a_k \cdot e^{-j\omega t_k}$ (linearity property).

3. We know that, as a signal:

$e^{-at} u(t) \xleftrightarrow{F} \frac{1}{a + j\omega}, \quad a > 0.$

$\therefore$ As a distribution too:

$e^{-at} u(t) \xleftrightarrow{F} \frac{1}{a + j\omega}.$

Use duality:

$\frac{1}{a + jt} \xleftrightarrow{F} 2\pi \cdot e^{-a(-\omega)} \cdot u(-\omega)$
 $= 2\pi \cdot e^{a\omega} u(-\omega).$

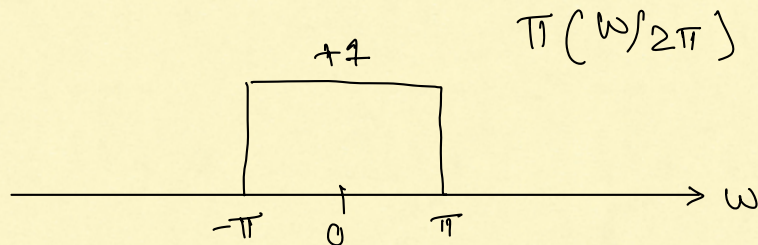
(as distribution)

4. Consider:

$$\sin\left(\frac{t}{2\pi}\right) \xleftrightarrow{F} 2\pi \cdot \Pi(\omega).$$

Apply time-scaling with $a = 2\pi$:

$$\sin(t) \xleftrightarrow{F} \frac{1}{2\pi} \cdot 2\pi \cdot \Pi(\omega/2\pi) = \Pi(\omega/2\pi).$$



5. We know that:

$$1 \xleftrightarrow{F} 2\pi \delta(\omega)$$

$$\Rightarrow -jt \cdot 1 \xleftrightarrow{F} 2\pi \delta'(\omega)$$

$$\Rightarrow t \xleftrightarrow{F} j2\pi \cdot \delta'(\omega).$$

6. We know that $\delta(t) \xleftrightarrow{F} 1$

$$\delta'(t) \xleftrightarrow{F} j\omega \cdot 1 = j\omega$$